

WASP-12b

R-1803

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160212 · created 2026-08-03T02:30:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457431.79943 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.044
Transit depth (Rp/Rs)^2	2.39 +/- 1.38 [%]
Orbital Inclination (inc)	78.6 +/- 2.84
Airmass coefficient 1 (a1)	1.26 +/- 0.76
Airmass coefficient 2 (a2)	-0.115 +/- 0.095
Scatter in the residuals of the lightcurve fit is	61.67 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	7.5
Transit Duration (day)	0.116 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.044 vs 0.1178 (deviation 0.68σ)
Duration fit vs published	0.116 d vs 0.125 d (deviation 0.55σ)
Mid-time O–C	1571.4 min
Depth SNR	2.8
Residual scatter	61.67 %

Light curve

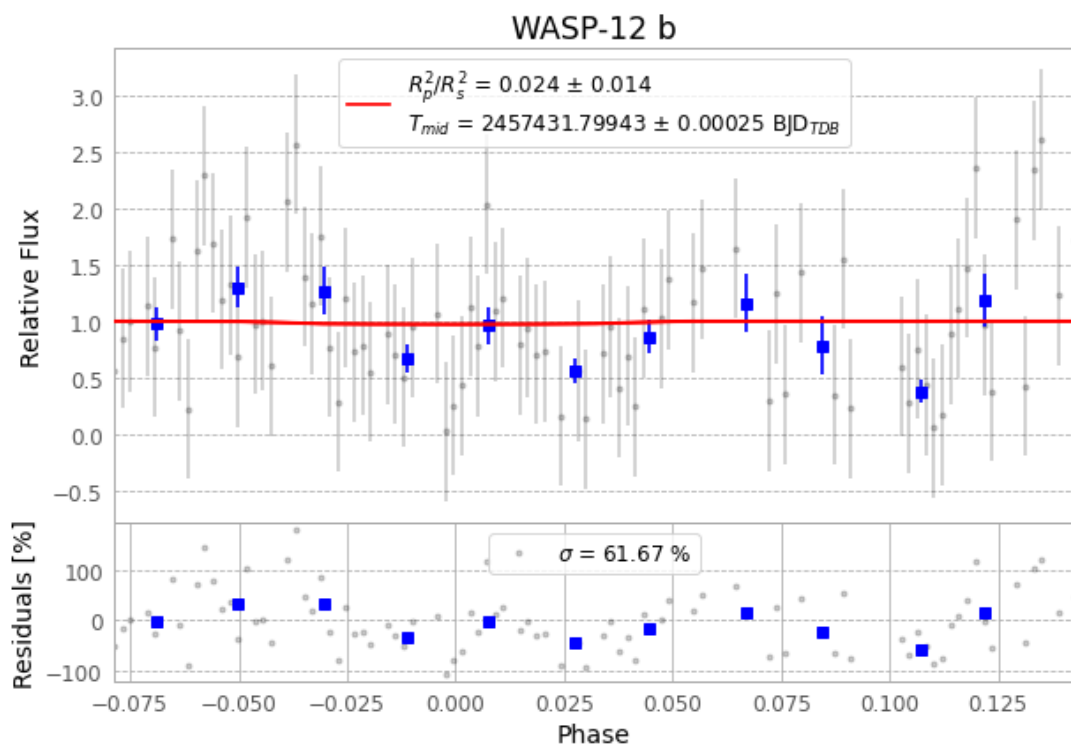


Figure 1 — FinalLightCurve_WASP-12 b_2016-02-12.png (SHA-256 466fd5de50ee806e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [582, 410], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a56f7b0046a5de41...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

117 FITS frames (77 MB), WASP-12160213050012.FITS → WASP-12160213104812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160212.log (758c5e43dae3...), FinalLightCurve_WASP-12 b_2016-02-12.png (466fd5de50ee...), FinalLightCurve_WASP-12 b_2016-02-12.csv (41e06cd3ba96...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 02:30Z.